Deflection of Beams

Objective

To study the deflection of a simply supported beam and compare the experimental values of deflection with the theoretical values.

Equipments

Dial gauges, meter scale, vernier caliper, weights, weighing pan, aluminum and mild steel beams of rectangular cross section, displacement sensor and load cell.

Theory

A beam is a slender member subjected to transverse loads. The beam bends to resist the loads applied. When transverse loads bend a beam, in general there will be both bending moment and shear force acting on each cross section of the beam. Both of these contribute to the deflection of the beam. When the beam deflection is measured experimentally one records the net effect.

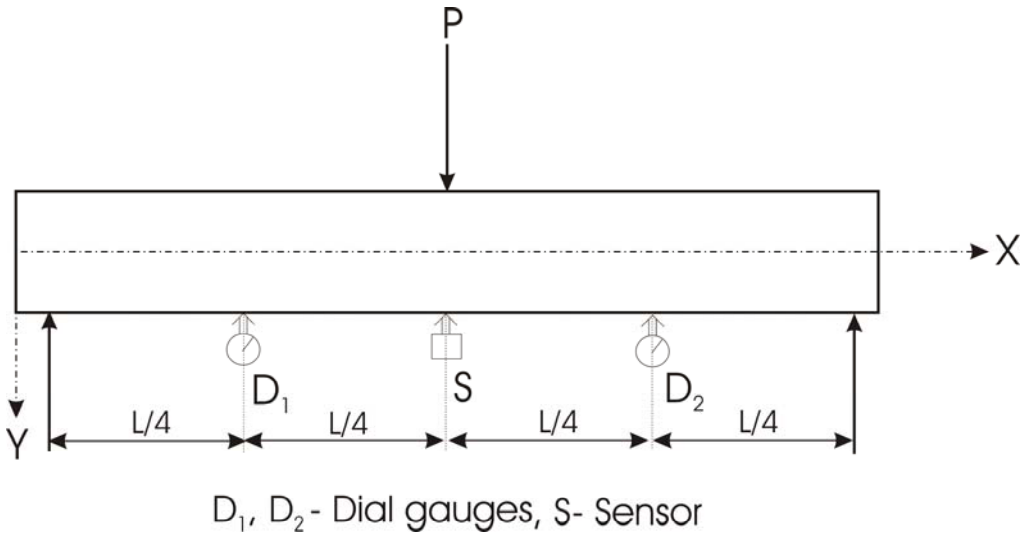


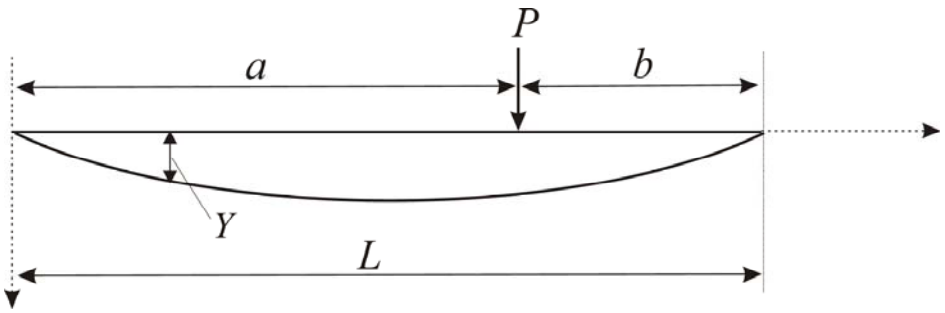
Fig.1. Experimental apparatus for beam deflection

Analytical Expressions for Deflection

For analytical computations since the thickness of the beam is very small compared to other dimensions; the effect of shear forces is usually neglected. This is done to simplify the mathematics. It is well known that for a simply supported beam subjected to a load of P at the centre, the deflection at the centre neglecting the effects due to shear is

$$Y_{\max} = \frac{PL^3}{48EI} \tag{1}$$

Deflection at any point of a simply supported beam with the loading at an arbitrary position can be calculated using the formula.



$$Y = \frac{Pbx(L^2 - b^2 - x^2)}{6LEI} \qquad (0 \le x \le a) \tag{2}$$

$$Y = \frac{Pbx(L^2 - b^2 - x^2)}{6LEI} + \frac{P(x-a)^3}{6EI} \qquad (a \le x \le L) \tag{3}$$

Where, *E* is Young's modulus, *I* is Second moment of area, *y* is Deflection, *P* is the Load applied and *L* is Length of the beam.

Procedure

1. Place the beam over the supports at a known distance apart.
2. Place the dial gauges at the centre, on either side of the beam at a distance 'L/4' from the center of the beam. Adjust the dial gauges such that the tip of the dial gauge remains in contact with the beam even when loaded.
3. The beam usually has an initial deflection due to its self-weight and the dial gauges readings are to be set to zero value.
4. Move the load cell exactly at the middle of the beam. Rotate the knob of load cell which apply the load on the test sample. Note down the applied load on the indicator. Also, note down the deflection of the beam by means of the dial gauges and displacement sensors at three different points.
5. Load in steps of 500gm weight until it reaches 2500gm. For each loading, note the readings in the dial gauges and displacement sensors.

6. Follow a similar procedure during unloading of the member and note the readings.
7. Now place the load cell at ‘L/4’ from one end and place the dial gauges at ‘L/2’ and at ‘3L/4’ from one end of the beam.
8. Follow the above procedure for this case also and note the readings.
9. Now place the load cell at ‘3L/4’ from one end and place the dial gages under the load at ‘L/4’ and at ‘L/2’ from one end of the beam.
10. Repeat the experiment for the aluminum specimen.

Sample Table

Location of loading: Specimen material:

Specimen dimensions: Length: Thickness: Width:

S. No.	Load applied		Deflection at (L/4)					Deflection at (3L/4)				
	gm	N	Loading		Unloading		Avg. mm	Loading		Unloading		Avg. mm
			div.	mm	div.	mm		div	mm	div	mm	
1												
2												
3												
4												
5												

Deflection at load application point				
Loading		Unloading		Avg. mm
div	mm	div	mm	

Analysis of Results

1. Compare the experimental and theoretical values of deflection. Present it in the form of a graph.

2. Evaluate the stiffness of both the aluminum and mild steel beam. Comment on their comparison.

3. Discuss the reason for differences between the experimental and theoretical values of deflections.

Points for Discussion

1. Can you verify the validity of the formulae shown in Eq.(1) and (2)? Determine the percentage of deviation of the experimental results with theory. Tabulate your results.

2. Does the overhang in the beam affect the deflection measured?

3. Among, aluminum or mild steel specimen of some geometric dimensions which will deflect more?

4. Does moment of inertia affect the deflection?

5. Does self-weight of the beam play a role on the deflection? How will you modify the procedure to account for this in the experiment?

6. Is the deflection symmetric when the load is placed at the centre?

7. When the loading is asymmetric do you find any interesting relationship between the loading and the deflection? Comment*.